

DO NOT WRITE ON TEST



**COMPUTER
PROGRAMMING
CONCEPTS
~OPEN EVENT~**

(390)

STATE 2021

TOTAL POINTS _____ **100 Points**

Test Time: 60 minutes

Choose the *best* answer to the following questions. Mark A if the statement is true. Mark B if the statement is false.

1. What program translates code in one language to code in another language all at once?
 - A. Interpreter
 - B. Executor
 - C. Compiler
 - D. Translator

2. The form of a program are dictated by this set of rules.
 - A. semantics
 - B. syntax
 - C. pseudocode
 - D. algorithm

3. Predefined identifiers/words that have a special meaning in a programming language are called _____.
 - A. Comments
 - B. Reserved words
 - C. String literals
 - D. Syntax

4. What is I/O?
 - A. Input and Output
 - B. Inter-Operations
 - C. Ignore and Operations
 - D. Independent and Operators

5. What dictates the meaning of programming statements?
 - A. syntax
 - B. semantics
 - C. pseudocode
 - D. algorithm

6. At least one of the conditions, connected by the _____ operator, must be true for the compound condition to be true.
 - A. OR
 - B. NOT
 - C. AND
 - D. NULL

7. How many steps are there in the program development cycle?
 - A. Ten
 - B. Two
 - C. Four
 - D. Seven
8. What is the basic order of operations as it pertains to programming?
 - A. Multiplication, Division, Parentheses, Exponents, Addition, Subtraction
 - B. Parentheses, Multiplication, Exponents, Division, Addition, Subtraction
 - C. Parentheses, Exponents, Multiplication, Division, Addition, Subtraction
 - D. Multiplication, Division, Modulus, Addition, Subtraction
9. What is it called when a data string appears in the actual code of a program?
 - A. string
 - B. string literal
 - C. an error
 - D. character
10. A mistake that prevents the program from running beyond the point of the mistake is known as a(n) _____.
 - A. syntax error
 - B. run-time error
 - C. logic error
 - D. fatal error
11. What is the term for having a single method with multiple implementations?
 - A. overusing
 - B. overloading
 - C. overriding
 - D. overcomplicated
12. What is the term for having multiple methods with the same name in a class?
 - A. overusing
 - B. overriding
 - C. overcomplicated
 - D. overloading
13. What do you use to convert from one data type to another?
 - A. casting
 - B. converting
 - C. changing
 - D. choosing

14. When objects react differently when the same method is called it is known as _____.

- A. encapsulation
- B. converting
- C. polymorphism
- D. endomorphism

15. The grouping of data and behavior together into distinct bundles is known as _____.

- A. polymorphism
- B. encapsulation
- C. endomorphism
- D. stacking

16. What will the following snippet of code do?

```
int counter = 1;
int total = 0;
while(counter <= 10)
{
    total += counter;
    counter++;
}
```

- A. Create an infinite loop.
- B. Print all of the numbers from 1 to 10.
- C. Print all of the numbers from 0 to 10.
- D. Calculate all of the numbers from 1 to 10.

17. What will happen if you assign a value higher than what the variable can hold?

- A. syntax error
- B. overflow error
- C. logic error
- D. run-time error

18. What does a constructor do?

- A. Creates objects based on the class datatype.
- B. Designates objects based on the class datatype.
- C. Designs objects based on the class datatype.
- D. Executes objects based on the class datatype.

19. What does `clear()` accomplish?
- A. changes the background to transparent
 - B. removes all text from the screen
 - C. removes all objects from the list
 - D. deletes the program
20. A procedure changes one or more attribute values.
- A. True
 - B. False
21. What is it called when you capitalize the first letter in the procedure name and the first letter of each subsequent word in the name?
- A. Camel case
 - B. Upper case
 - C. Superior case
 - D. Pascal case
22. Which expression is executed first in a *for* loop?
- A. for loop statement
 - B. initial expression
 - C. logical expression
 - D. update expression
23. What is the value of the following expression: $5 * 3 / 3 > 6 / 2 * 5$
- A. 5
 - B. True
 - C. False
 - D. 15
24. The following set of statements represents what type of structure?
- ```
if temperature is greater than sixty
 print it is a beautiful day out
else
 print it is a little chilly out, wear a jacket
end if
```
- A. selection
  - B. sequence
  - C. iterative
  - D. collective

25. What is the process of locating and correcting errors known as?
- A. debugging
  - B. defragging
  - C. degaussing
  - D. derailment
26. Flow control refers to the sequence in which a computer executes program instructions.
- A. True
  - B. False
27. You can limit the complexity of a problem statement by making \_\_\_\_\_.
- A. limits
  - B. assumptions
  - C. tests
  - D. errors
28. A graphical representation of how a program progresses from one instruction to the next is known as \_\_\_\_\_.
- A. pseudocode
  - B. development cycle
  - C. graphical depiction
  - D. flowchart
29. Programmers can ignore the details of machine language if they work with \_\_\_\_\_.
- A. low-level language
  - B. machine language
  - C. high-level language
  - D. assembly language
30. When a program structure repeats a block of statements, this is known as \_\_\_\_\_.
- A. Infinite
  - B. Sequence
  - C. Selection
  - D. Iterative
31. Each programmer writes pseudocode the same.
- A. True
  - B. False

32. Which data type can contain real numbers?
- A. float
  - B. char
  - C. bool
  - D. int
33. What type of values do comparison operators evaluate?
- A. integer
  - B. string
  - C. character
  - D. Boolean
34. What is the unique number, identifying each item in a collection, called?
- A. counter
  - B. unique number
  - C. index
  - D. identifier
35. What is the numeric variable that is used to add something together called?
- A. index
  - B. identifier
  - C. increment
  - D. accumulator
36. In a flowchart, what is the symbol used for calculations?
- A. Rectangle
  - B. Oval
  - C. Diamond
  - D. Parallelogram
37. Given the decimal number 13, what is the binary equivalent?
- A. 1001
  - B. 1101
  - C. 0111
  - D. 1111
38. What is the correct precedence for logical operators?
- A. AND, NOT, OR
  - B. NOT, AND, OR
  - C. NOT, OR, AND
  - D. OR, AND, NOT

39. How are the variables in a class defined?
- A. by function
  - B. by class methods
  - C. by control structure
  - D. by class attributes
40. Which is considered a machine language instruction?
- A. goto 100
  - B. printf("welcome")
  - C. 110001 000100 111001 0011110 000011 100100
  - D. mov ax, 0002
41. The user can typically make selections using a mouse or other pointing device in a \_\_\_\_\_ program.
- A. CLI
  - B. DOS
  - C. GUI
  - D. VR
42. When naming a variable, which rule can apply?
- A. Variable names are allowed to also be keywords.
  - B. Variable names may contain an underscore.
  - C. Variable names may begin with numbers.
  - D. Variable names may contain spaces.
43. Desk-checking is also known as \_\_\_\_\_.
- A. hand-tracing
  - B. cross checking
  - C. design checking
  - D. developing
44. Memory addressing is often referred to as \_\_\_\_\_ notation or base 16.
- A. Hexadecimal
  - B. Binary
  - C. Mathematical
  - D. Scientific
45. \_\_\_\_\_ is a type of error caused by factors during the execution of a program.
- A. Logic error
  - B. Syntax error
  - C. Run-time error
  - D. Compiler error

46. Multi-line commenting start with which characters?
- A. //
  - B. “
  - C. ‘
  - D. /\*
47. In a nested loop, the inner loop goes through all of its iterations for every iteration of the outer loop.
- A. True
  - B. False
48. When determining if a number is inside a range, the programmer should use which operator?
- A. !
  - B. ||
  - C. &&
  - D. ==
49. Which Scanner class method would you use to read a double as input?
- A. scanDouble
  - B. nextDouble
  - C. nextLine
  - D. scanLine
50. The output of the following code segment is \_\_\_\_\_.
- ```
int x = 5;
System.out.println(x++);
```
- A. 6
 - B. 7
 - C. 5
 - D. 0